

Homework 2

Handed Out: 2/3/2026

Due: 2/14/2024

1. (10 points) Recall FTRL with entropy regularization for OCO with linear loss:

$$x_t = \operatorname{argmin}_{x \in V} \frac{1}{\eta} \sum_{i \in [d]} x_i \log x_i + \sum_{i=1}^{t-1} \ell_t(x).$$

Follow Lemma 16 in [1], $\psi(x) = \sum_{i \in [d]} x_i \log x_i$ is 1-strongly convex in 1-norm. Prove the expected regret of FTRL with entropy regularization with your choice of η .

2. (10 points) Consider the expert problem. Show that for any deterministic policy, there exists an environment such that the regret is linear to T .
3. (30 points) Assume that X_1 and X_2 are independent and σ_1 -subgaussian and σ_2 -subgaussian respectively. Recall that σ -subgaussian random variable is defined as

$$\mathbb{E}[\exp(\lambda X)] \leq \exp(\sigma^2 \lambda^2 / 2)$$

- (1) (10 points) [Chernoff bound] Prove

$$\mathbb{P}(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$$

[Hint: use Markov's inequality $\mathbb{P}(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon}$]

- (2) (10 points) Prove that cX_1 is $|c|\sigma_1$ -subgaussian, $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subgaussian.
- (3) (10 points) [Hoeffding's inequality] Prove that for any random variable X such that $X \in [a, b]$ with independent samples $X_i \in [a_i, b_i]$,

$$\mathbb{P}\left(\frac{1}{n} \sum_{t=1}^n (X_t - \mathbb{E}[X]) \geq \epsilon\right) \leq \exp\left(-\frac{2n^2\epsilon^2}{\sum_{t=1}^n (b_t - a_t)^2}\right)$$

4. (20 points) Prove the following gap-independent regret bound of UCB algorithm.

Theorem 1 Assume that the rewards of the arms minus their expectations are 1-subgaussian and let $\alpha > 2$. Then, UCB guarantees a regret of

$$\operatorname{Regret}_T \leq 4\sqrt{2\alpha d T \ln T} + \frac{\alpha}{\alpha - 2} \sum_{i=1}^d \Delta_i.$$

[Hint: separate regret Δ_i by a threshold.]

5. (30 points) Consider ϵ -greedy algorithm with decaying exploration probabilities for d -armed bandits with rewards $r \in [0, 1]$.

Algorithm 1 Epsilon-Greedy with exploration probabilities $(\epsilon_1, \epsilon_2, \dots)$.

1. **for** each round $t = 1, 2, \dots, T$ **do**
 2. With probability ϵ_t , explore: choose an arm uniformly at random
 3. Otherwise, exploit: choose the arm $A_t = \operatorname{argmax}_i \hat{\mu}_{t-1,i}$
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Show that $\epsilon_t = t^{-1/3}(d \log T)^{1/3}$ gives a $O(T^{2/3}(d \log T)^{1/3})$ gap-independent regret bound.

[Hint] Rewards $r \in [0, 1]$ means the reward is a 1-subgaussian random variable. $S_{t,i}$ is the number of rounds we choose arm i until t . Apply the concentration inequality in Lecture 8 (also known as Hoeffding's inequality, you do not need to prove this)

$$\mathbb{P}(|\hat{\mu}_{t,i} - \mu_i| > c_t) \leq 2 \exp(-2S_{t,i}c_t^2)$$

. Let the probability on the right hand side be $2/T^4$, we have the confidence radius $c_t = ?$

Define clean events as

$$C = \{|\hat{\mu}_{t,i} - \mu_i| \leq c_t, \forall i \in [d]\}$$

, by union bound show

$$\mathbb{P}(\bar{C}) = \mathbb{P}(\{\exists i, |\hat{\mu}_{t,i} - \mu_i| > c_t\}) \leq ?$$

Let $A^* = \operatorname{argmax}_i \mu_i$. You should prove that regret at time t is

$$\begin{aligned} \mu_{A^*} - \mu_{A_t} &\leq \epsilon_t \cdot 1 + (1 - \epsilon_t)(\mu_{A^*} - \mu_{A_t}) \\ &\leq \epsilon_t + (1 - \mathbb{P}(\bar{C}))2c_t + \mathbb{P}(\bar{C}) \cdot 1 \\ &\leq \epsilon_t + 2c_t + \mathbb{P}(\bar{C}) \end{aligned}$$

Regret can be decomposed as

$$\begin{aligned} R_T &= \sum_{t=1}^T (\mu_{A^*} - \mu_{A_t}) \\ &\leq \sum_{t=1}^T (\epsilon_t + 2c_t + \mathbb{P}(\bar{C})) \end{aligned}$$

Finally, by setting $\epsilon_t = t^{-1/3}(d \log T)^{1/3}$, you can prove the regret.

6. (Bonus, 10 points) Assume that $d = 2$ and let $\delta \in (0, 1)$. Modify the ETC algorithm to depend on δ and prove a gap-dependent regret bound of ETC holds with probability $1 - \delta$, i.e., a high probability bound.

[Hint] Now choose m related to δ , e.g., $m = \min\{T/d, \frac{4}{\Delta^2} \log(\frac{1}{\delta})\}$.

References

- [1] S. Shalev-Shwartz. *Online learning: Theory, algorithms, and applications*. Hebrew University, 2007.